

CY 507: Chemistry of Main Group Elements

Course Code	CY 507
Course Name	Chemistry of Main Group Elements
L-T-P-C	3-1-0-4
Prerequisites	B.Sc. (with Chemistry) or Teachers consent
Intended for	UG/PG
Distribution	Core
Approval	8th Senate (Oct 2015)

Course Contents

- **Module I**
Chemical periodicity and periodic anomalies. [5 Lectures]
- **Module II**
Structure and bonding in homo- and heteronuclear molecules, shapes of small molecules. [10 Lectures]
- **Module III**
Concepts of acids and bases: Bronsted and Lewis acids and bases - Gas phase versus solution acidity - leveling effects of solvents - hardness and softness - surface acidity. [10 Lectures]
- **Module IV**
Pre-transition (alkali and alkaline earth) metals, their solutions in liquid ammonia. Noble gas compounds, structure and reactivity. Hydrogen and its compounds, H-bond and its influence on the structure and properties of crystals. [10 Lectures]
- **Module V**
Chemistry of main group elements and their compounds: Borides, borates, boron halides, boranes, carboranes and metallocarboranes. BN compounds - a brief survey of Al, Ga and In chemistry. Allotropes of carbon - intercalation compounds carbides - C-oxides - pure silicon – silica and silicates - silicones low coordinated silicon compounds a brief survey of Ge, Sn chemistry. P(III) and P(V) compounds, P-N, P-O and P-S compounds including cyclophosphazenes a brief survey of heavier elements - nitrogen and phosphorus ligands - Sulfur-nitrogen compounds. Charge-transfer complexes of halogens Inter-halogen compounds - halogen oxides and oxygen fluorides - pseudohalogens. [20 Lectures]

Text and Reference Books

1. F. A. Cotton, G. Wilkinson, C. A. Murillo and M. Bochmann, **Advanced Inorganic Chemistry**, 6th Edition, Wiley 2007.
2. J. E. Huheey, E. A. Keiter and R. L. Keiter, **Inorganic Chemistry: Principles of Structure and Reactivity**, 4th Edition, Pearson Education Inc., 2000.
3. B. Douglas, D. McDaniel and J. Alexander, **Concepts and Models of Inorganic Chemistry**, 3rd Edition, Wiley, 2006.
4. K. F. Purcell and J. C. Kotz, **Inorganic Chemistry**, Cengage Learning, 2010.
5. A. G. Sharpe, **Inorganic Chemistry**, 3rd Edition, Pearson, 2011.
6. P. Atkins et al, **Shriver & Atkins Inorganic Chemistry**, 5th Edition, W. H. Freeman and Company, 2010.